

One other note I want to make is that if $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultra-net and $X = \bigcup_{i=1}^n E_i$, then we can conclude that there is at least one E_i such that x_α is eventually in E_i .

Proof:

We proceed by induction on n . For our base case, suppose $X = E_1 \cup E_2$. Then if x_α is not eventually in E_1 , we know that x_α is eventually in $X - E_1$. And since $X - E_1 \subseteq E_2$, we can conclude that x_α is eventually in E_2 .

Now suppose $n > 2$ and that $X = \bigcup_{i=1}^n E_i$. Then by writing:

$$X = (E_1 \cup E_2) \cup E_3 \cup \dots \cup E_n,$$

we can conclude by induction that either there exists $i \geq 3$ with x_α eventually in E_i , or we have that x_α is eventually in $E_1 \cup E_2$. We're trivially done in the former case. So assume the latter case is what happens.

Next, write $X = E_1 \cup (E_2 \cup E_3) \cup E_4 \cup \dots \cup E_n$. Then we know from our induction hypothesis that either there exists $i \neq 2, 3$ with x_α eventually in E_i or we have that x_α is eventually in $E_2 \cup E_3$. We assume the former case doesn't happen since otherwise we'd be trivially done. In particular, this means that x_α is not eventually in E_1 .

But now we have that x_α is eventually in $(E_1 \cup E_2)$, and yet at the same time since $\langle x_\alpha \rangle_{\alpha \in A}$ is an ultrafilter, we can conclude that x_α is eventually in $X - E_1$. In turn, we can conclude that x_α is eventually in $(E_1 \cup E_2) \cap (X - E_1) = E_2 - E_1 \subseteq E_2$. ■

Proposition: Suppose X is a uniform space. Then X is totally bounded if and only if every net $\langle x_\alpha \rangle_{\alpha \in A}$ in X admits a Cauchy subnet.

($\Rightarrow$)

Let $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ be an ultra-subnet of $\langle x_\alpha \rangle_{\alpha \in A}$. Then I claim that $\langle x_{\alpha_\beta} \rangle_{\beta \in B}$ is Cauchy since more generally, any ultra-net $\langle y_\gamma \rangle_{\gamma \in C}$ is Cauchy in a totally bounded space.

To see why, let $\{y_\gamma\}_{\gamma \in C}$ be an ultra-net. Then suppose E is any entourage of X . Firstly, we find an entourage E' such that $E' = (E')^{-1}$ and $E' \circ E' \subseteq E$. Next, note that since X is totally bounded, there exists $x_1, \dots, x_n \in X$ such that $X = \bigcup_{i=1}^n E'[x_i]$. Since $\{y_\gamma\}_{\gamma \in C}$ is an ultra-net, we know there must exist some x_i such that y_γ is eventually in $E'[x_i]$.

But now let $\gamma_0 \in C$ be such that $y_\gamma \in E'[x_i]$ for all $\gamma \geq \gamma_0$. Then for all $\gamma_1, \gamma_2 \geq \gamma_0$ in C we have that $x_i E' y_{\gamma_1}$ and $x_i E' y_{\gamma_2}$. This implies that $y_{\gamma_1} E y_{\gamma_2}$.

($\Leftarrow$)

Suppose that X is not a totally bounded space. Thus, we know there exists an entourage E such that $E[S]$ is a proper subset of X for all finite subsets S of X .

Let $\mathcal{S}$ be the collection of all finite subsets of X . This becomes a directed set when partially ordered by inclusion. Next, for every $S \in \mathcal{S}$ let x_S be an element of $X - E[S]$. Then, we can consider the net $\langle x_S \rangle_{S \in \mathcal{S}}$.

I claim that $\langle x_S \rangle_{S \in \mathcal{S}}$ has no Cauchy subnets. To see why, suppose that $\langle x_{S_\beta} \rangle_{\beta \in B}$ is any subnet of $\langle x_S \rangle_{S \in \mathcal{S}}$. Then consider any fixed $\beta_0 \in B$. It suffices to show that there exists $\beta_1, \beta_2 \geq \beta_0$ in B such that $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. Fortunately, just let $\beta_1 = \beta_0$ and then let $\beta_2 \in B$ be such that $\beta_2 \geq \beta_0$ and $S_{\beta_2} \supseteq S_{\beta_0} \cup \{x_{S_{\beta_0}}\}$. Then we know $(x_{S_{\beta_1}}, x_{S_{\beta_2}}) \notin E$. ■

Before wrapping up for the day, I also want to translate the prior result into a result about Cauchy sequences in the case that the uniform structure on X has a countable filterbase.